X-UAD: A Unified, Uncertainty-Aware and Self-Explaining Detector for Physical and Digital Face Attacks

Abstract—Face recognition systems can be defeated by presentation attacks staged in front of the camera, such as a printed photograph, a replayed video or a silicone mask, and by digital manipulation of the captured image through face swapping, expression reenactment, synthesis or adversarial perturbation. A detector built for one family is usually blind to the other, and the recent unified detectors that handle both still share three practical weaknesses. They emit a single uncalibrated score with no way of signalling that an input resembles nothing they were trained on; they cannot point to where in the frame the evidence sits; and when they do explain themselves, the explanation comes from a language model that never observed the detector's internal state. We present X-UAD, one model aimed at all three. Adaptive Quantum Attention adds a Born-rule scoring channel beside ordinary dot-product attention. It is sign-invariant by construction, which matches the interference character of moiré, halftone and checkerboard evidence, and it costs $\mathcal{O}(dL)$ parameters rather than $\mathcal{O}(d^2)$. The same learnable rotations define an orthonormal evidence basis whose measurements serve simultaneously as classification features and, completely unchanged, as the explanation shown to a reader. A decoupled evidential head predicts class direction and evidence magnitude separately, and pseudo-unseen evidence suppression trains that magnitude on blends of two attack families, an outlier proxy built from the attack taxonomy itself. On a difficulty-calibrated, identity-disjoint benchmark of eight attack families with pixel-level ground truth, X-UAD reaches 0.42% ACER on seen attacks. That is statistically level with the strongest baseline, CDCN++ (0.46%), and well ahead of ResNet-18 (1.88%), although X-UAD trails both on eight-way family accuracy (94.1% against 95.2% and 96.9%). It localises the evidence best, at 80.1% region-level IoU over the full test set against 73.2% for CDCN++. On attack families withheld from training it has the highest average detection rate (77.5%), and the synthetic-blend objective is worth 5.4 AUROC points on the uncertainty channel. The results have clear limits: every method collapses under a change of capture domain, a white-box attacker can suppress X-UAD's uncertainty without changing its decision, and all numbers come from a single seed on a simulated benchmark. Code, the benchmark generator and configuration files mapping the same pipeline onto public capture-based datasets are released.

Index Terms—face anti-spoofing, presentation attack detection, deepfake detection, unified attack detection, vision transformer, quantum-inspired attention, evidential deep learning, uncertainty estimation, open-set recognition, explainable AI

I. Introduction

Consider everything that can go wrong in front of a face-unlock camera. Someone could hold up a printed photo of you. They could replay a video of your face on a phone screen. They could wear a silicone mask shaped like your face. These are physical attacks, and each leaves a physical fingerprint behind: the halftone dot pattern of a printer, the pixel grid of a phone screen aliasing into a moiré pattern, the waxy sheen and flattened texture of silicone.

Now consider what can go wrong after the camera has captured a real, unmanipulated photograph. Someone could swap in a different identity. They could drive your expression with someone else's performance. They could generate a face that never belonged to any real person. They could add a barely visible pattern of noise that flips a classifier's decision without a human noticing anything at all. These are digital attacks, and their fingerprints look entirely different: blending seams where a swapped face meets the original, resolution mismatches, checkerboard patterns left behind by the upsampling layers of a generator, unusual peaks in the frequency spectrum.

Because these two sets of fingerprints look nothing alike, the two research communities that study them grew up separately. Face anti-spoofing work focused on texture and depth cues [1]–[4]. Face forgery research focused on frequency artifacts and blending cues [5]–[8]. The split is understandable, but it is not something you can deploy. An operator needs one decision, one threshold and one audit trail, not two models that might disagree with no principled way to combine them. A recent line of work [9]–[13] has finally built unified corpora and joint detectors that handle both attack families at once, which is real progress. It also exposed a problem that had been hiding underneath. On the attack types these models were trained on they are essentially saturated, with ACER around 0.5 to 1.4%, but the moment you hold out an attack family during training and show it only at test time, error rates jump to between 22 and 44% with large variance between runs [9]. Unified detection is close to solved in-distribution and still wide open out-of-distribution.

Layered on top of that gap, we see three capabilities missing from almost all of this literature, and an operator needs all three the moment a score comes back ambiguous rather than obviously fake or obviously real.

The first is a real notion of not knowing. A softmax output is fundamentally a closed-set quantity, since it has to redistribute all of its probability mass across the classes it was trained on. A completely novel kind of attack therefore gets assigned high confidence toward whichever

known class it happens to resemble most. What is needed is a second-order belief, one that is allowed to sit close to empty when the input does not look like anything the model has learned.

The second is evidence localisation. Depth-map supervision [1], [3] works reasonably for physical attacks, since a printed photo or a replayed video really is flatter than a real face. But depth is a proxy for liveness rather than a map of forensic evidence, and it says nothing useful about a digitally swapped face that came from a perfectly live, perfectly three-dimensional original capture.

The third is an explanation that is tied to the decision. Recent work has started feeding face images into multimodal language models and asking them to describe what looks fake [14], [15]. The results read fluently, but the language model never sees the detector's internal state. It is reasoning over the raw image, independently of whatever the classifier computed. Its narrative is therefore correlated with the decision at best, and it is free to mention artifacts such as "a visible screen bezel" that are not present in the frame or that the detector never used.

A. What this paper does

We built X-UAD, a single model that detects physical and digital face attacks together, reports how much it does not know, points to where the evidence is, and explains its own decision in language that traces back to numbers it computed. Three ideas make this work.

Adaptive Quantum Attention (AQA). Ordinary attention scores a pair of tokens with an unbounded linear functional, the dot product. A great deal of the evidence for face attacks, though, is fundamentally an interference phenomenon: a moiré beat, a checkerboard pattern, a spectral peak. These show up in the squared magnitude of a projection, not in the raw projection, and their sign carries no information. A moiré fringe and its photographic negative are the same evidence, but a dot product treats them as opposites. AQA adds a second scoring channel based on the Born rule from quantum measurement. Each token is represented as a unit vector, a "pure state", evolved by a small stack of learnable rotation matrices (a Givens circuit), and two tokens are compared by the squared overlap between their evolved states. This channel is mixed into the ordinary channel through a learned, per-head, input-conditioned gate. The rotation matrices are exactly orthogonal by construction and are parameterised by $\mathcal{O}(dL)$ angles rather than the $\mathcal{O}(d^2)$ weights of a normal linear layer.

A quantum evidence basis that makes the explanation part of the computation. A learnable orthonormal set of basis vectors, built as the image of the standard basis under a rotation circuit, stays exactly orthonormal at every step of training and is measured against every spatial token. The resulting probabilities form a token-by-mode evidence map that is used as a classification feature and, completely unchanged, as the explanation. The explanation is therefore a direct read-out of the computation rather than a post-hoc approximation of it produced by a separate model.

Pseudo-Unseen Evidence Suppression (PUES). If you take two training images from two different attack families and blend them, you get something no real camera or generator could ever produce, because no physical process makes a superposition of a print attack and a face swap. That makes these blends a free, built-in proxy for "novel input". We train the model to push its evidential output toward zero on these blends, which teaches the uncertainty channel what "I do not recognise this" should look like without ever showing the model a real unseen attack.

We also built evidence-grounded reasoning. Every forward pass emits a typed, numeric evidence record, and the natural-language explanation is generated only from that record, either by a deterministic rule-based generator whose hallucination rate is zero by construction, since every sentence is gated by a numeric condition, or optionally by a hosted multimodal language model whose output is then automatically checked against the record.

Finally, because no existing capture-based dataset marks where in the frame the attack evidence lives, we built a difficulty-calibrated, identity-disjoint benchmark of eight attack families with pixel-level evidence masks, along with identity-consistent, leave-one-attack-out and cross-domain protocols, and an explicit calibration procedure so the benchmark cannot be trivially solved.

Section II surveys the research threads X-UAD tries to bring together and states the gap explicitly. Section III describes the benchmark. Section IV describes the model. Section V reports the experiments. Section VI returns to the gap and explains, piece by piece, what closes it.

II. Related Work and the Gap

A. Physical presentation attack detection

Early face anti-spoofing treated liveness as a texture classification problem. Auxiliary supervision [1] replaced the plain real/fake label with a pseudo-depth map and an rPPG signal as extra training targets. Central Difference Convolution [2] modified the very first layers of the network to respond to illumination-invariant micro-texture by mixing in the local gradient response, and it remains one of the strongest lightweight backbones for physical attacks specifically. Pixel-wise supervision was revisited more systematically in [3]. Several works then tackled the harder problem of generalising across capture domains, meaning different cameras and different lighting, using adversarial domain generalisation [16], single-side domain generalisation [17], and style-shuffled feature separation [4]. All of this work shares one assumption: that the attack happens in front of the camera. None of it has any mechanism for a capture that is physically genuine but was manipulated afterward.

B. Digital forgery detection

FaceForensics++ [5] and Celeb-DF [6] are the benchmarks this line of work grew up around. The most transferable cue turned out to be spectral. Generative models leave periodic artifacts from their upsampling layers [18], which show up as distinctive peaks in the Fourier magnitude [7], and this is a large part of why fairly simple detectors generalise surprisingly well across different generator architectures [8]. The symmetric weakness applies here too. These detectors have no concept of a reacquisition medium and fail badly on printed or replayed attacks.

C. Unified physical-digital detection

UniAttackData [9] is arguably the most important recent contribution in this space. It is an identity-consistent corpus in which the same person appears both genuine and under every attack type, so that identity itself cannot become a shortcut, paired with a vision-language detector using teacher-student prompting. The CVPR'24 and ICCV'25 challenges built on this [10], [11], scaling to nearly 2,900 subjects and 54 attack types, and confirmed two things worth remembering: methods built on large pretrained multimodal backbones dominate the leaderboard, and single-modality CNNs get stuck above 20% ACER on the harder protocol. UniDefense [13] approaches the same goal through dual-space reconstruction. What none of these systems has is an uncertainty channel, an evidence map, or a genuine explanation. All that comes out is a score.

D. Uncertainty and open-set recognition

Maximum softmax probability [19] is a weak out-of-distribution signal for the reason described above, since softmax is a closed-set construction. Prior networks [20] and evidential deep learning [21], built on subjective logic [22], place a Dirichlet distribution over the simplex so that total evidence, and therefore vacuity, becomes something a model can output. Outlier exposure [23] showed that having even a rough proxy for "something else" during training is worth more than any purely intrinsic uncertainty score. Open-set single-domain generalisation has recently been explored for anti-spoofing on its own [24], but to our knowledge evidential uncertainty has not been combined with unified physical-digital detection before, and no prior work builds its outlier proxy directly out of the attack taxonomy the way PUES does.

E. Explainability and MLLM-based forensics

Attention rollout [25] and deletion/insertion faithfulness testing [26] are the standard tools for checking whether an explanation reflects a model's decision. FFAA [14] fine-tunes a multimodal LLM for open-world forgery analysis with a multi-answer decision system, and M2F2-Det [15] bridges CLIP and an LLM for interpretable forged-face detection. Both produce fluent text, and both are, in a real sense, narrating after the fact. The language model sees the image but never the detector's internal evidence, so there is no way to check its account against the decision it is supposedly explaining, and it is free to describe artifacts that are not there.

F. Quantum-inspired attention

Quantum Vision Transformers [27] and quantum self-attention [28], [29] replace ordinary linear projections with parameterised quantum circuits, reporting accuracy comparable to classical vision transformers at a small fraction of the parameters, which is essentially an $\mathcal{O}(n^2) \to \mathcal{O}(n)$ argument. Quantum adaptive self-attention [30] places a parameterised circuit inside a single encoder layer and compares it against capacity-matched classical controls, to separate a quantum effect from an architectural one. This line of work is motivated primarily by parameter efficiency and is evaluated on generic image classification. It has not been connected to a task whose evidence is literally an interference pattern, and none of it offers an explainability story.

G. Stating the gap

Every one of the six lines above occupies a proper subset of what an operator needs (Table I), and the missing pieces are not just additive. Unifying the attack space creates the uncertainty problem, because more families means more ways to fall outside all of them. Adding uncertainty creates the explanation problem, because an operator asked to adjudicate a vacuous, low-confidence sample needs to know what the model saw. And an explanation is only worth trusting if it is tied to the same internal evidence that produced the decision in the first place. X-UAD is designed around this dependency chain rather than as a grab-bag union of separate modules bolted together.

III. The X-UAD Benchmark

A. Why we built our own benchmark

The capture-based corpora the unified-detection literature relies on [9], [11] are access-restricted, and more importantly for our purposes, none of them marks where in the frame the attack evidence sits. Training and scoring an evidence localiser needs ground truth that a real capture pipeline cannot provide, because nobody knows exactly which pixels a real printer or a real generator changed. We therefore build a benchmark where the evidence is injected by a known, documented operator, so its support is known exactly, by construction.

This involves a trade-off. A generated benchmark buys pixel-level supervision, perfect identity consistency, and full reproducibility, and it costs external validity, since an operator is a model of a physical or generative process rather than the process itself. We try to manage that risk in four ways. Every operator is motivated by the actual physics or generator architecture it stands in for. The benchmark is difficulty-calibrated so it cannot be solved

TABLE I
Capability matrix over the lines of work X-UAD unifies. Partial means the capability exists only for one attack super-class, or only as a proxy.

Representative work	Physical PAs	Digital attacks	Second-order uncertainty	Explicit unknown class	Evidence localisation	Natural-language explanation	Explanation bound to the decision
CDCN++ [2], pixel-wise FAS [3]	yes	no	no	no	partial (depth proxy)	no	—
SSAN [4], SSDG [17]	yes	no	no	no	no	no	—
Frequency forensics [7], [8]	no	yes	no	no	no	no	—
UniAttackDetection [9], UniDefense [13]	yes	yes	no	no	no	no	—
Challenge winners [11] (CLIP/VAE, LoRA ViT-L)	yes	yes	no	no	no	no	—
Evidential / open-set [21], [24]	partial	no	yes	yes	no	no	—
FFAA [14], M2F2-Det [15]	no	yes	no	partial	partial	yes	no (post-hoc)
Quantum ViT [27], [29], [30]	no	no	no	no	no	no	—
X-UAD (ours)	yes	yes	yes	yes	yes	yes	yes

trivially. We keep a second, different capture domain purely for cross-domain testing. And we release configuration files that map the identical taxonomy, protocols and model code onto CASIA-FASD [31], OULU-NPU [32], SiW-Mv2, CelebA-Spoof [33], FaceForensics++ [5], Celeb-DF-v2 [6] and UniAttackData [9], so that every claim in this paper is directly re-testable against real captured data by anyone who obtains access to those corpora. Nothing about the model, the losses, the training loop, the explainability code or the reasoning module is specific to our synthetic benchmark.

B. Source faces and identity discipline

Domain A (LFW-Sim) is built from 12,794 identity-labelled, tightly cropped face photographs of 5,616 subjects from Labeled Faces in the Wild [34]. Splits are assigned by a stable hash of the subject identity, so train, validation and test never share a subject. Every source face contributes exactly one genuine sample and one sample of each of the eight attack families, so the mutual information between identity and label is zero by construction and no detector can cheat by memorising who a person is rather than learning what an attack looks like. The source imagery is grayscale, and we treat that as a deliberate constraint rather than a limitation. It removes colour-based shortcuts such as display gamut shift, ink gamut compression and colour-transfer residue, and forces every method, ours included, to make its decision on texture, geometry and frequency evidence instead.

Domain B (Wild-Sim) applies the identical operators to 700 in-the-wild colour face crops at low and variable resolution with an unknown prior JPEG compression history. It is never used for training. Its only job is to be the target of the cross-domain protocol, and it differs from Domain A in colour, alignment, resolution and compression history all at once.

C. Attack taxonomy

Eight classes span the two super-classes (Fig. 1). Each operator takes a global strength parameter $s \in (0, 1]$ that scales every artifact's amplitude, and returns a soft mask $M \in [0, 1]^{H \times W}$ marking which pixels carry the injected evidence.

- print (physical): Yule–Nielsen tone transfer with dot gain, an ordered halftone screen at a sampled line frequency and angle, paper substrate grain, loss of live micro-texture, a small planar perspective shift, and a simulated re-acquisition chain comprising signal-dependent shot and read noise followed by JPEG compression.
- replay (physical): display gamut and contrast compression, an RGB sub-pixel stripe lattice at a sampled period and tilt that aliases into a visible low-frequency moiré beat, rolling-shutter banding, specular backlight glare, then re-acquisition and double compression.
- mask3d (physical): edge-preserving smoothing confined to the skin region, since silicone loses live micro-texture, with eye and mouth cut-outs through which the wearer's real features remain visible, a waxy albedo with broad specular sheen, and a seam at the mask's rim.
- faceswap (digital): a donor face aligned to the target's face box, passed through a deliberately low generator resolution and upsampled to mimic the resolution mismatch of a real swap pipeline, colour-transferred, and alpha-blended into the inner-face region. The residual evidence is the blending boundary and resolution mismatch, not gross misalignment.
- reenact (digital): a smooth non-rigid warp confined to the mouth and eye regions, with a local sharpness and contrast inconsistency inside the manipulated area only.
- synth (digital): the classic generator signature, namely over-smoothed skin combined with over-sharpened detail, a transposed-convolution checkerboard at a sampled stride [18], and an explicit azimuthally modulated radial peak injected into the Fourier magnitude [7].
- adversarial (digital): not random noise, but a genuine ℓ_∞ PGD perturbation ($\varepsilon = 6/255$, 7 steps) [35] crafted against an independently trained surrogate detector, a small VGG-style network that shares no weights, architecture or random seed with any model we evaluate, and is fit only on the training split. The perturbations seen at validation and test time

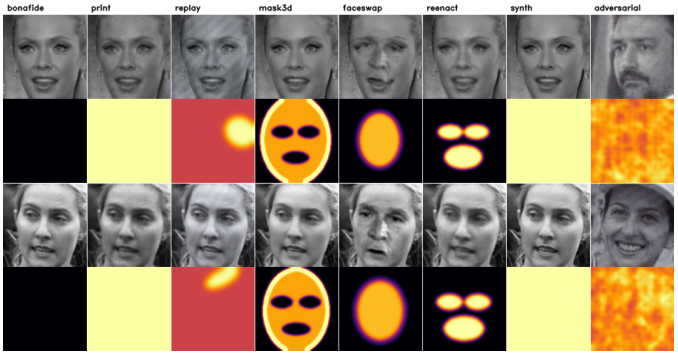


Fig. 1. The unified taxonomy. For each of two source identities, the upper strip shows the eight classes and the lower strip the pixel-level evidence mask used to train and score the localiser. Physical operators (print, replay) modify the whole capture; region-level operators (3-D mask, face-swap, reenactment, adversarial) have a localised support. The region-level IoU of Table VI is averaged over 3-D mask, face-swap and reenactment.

are therefore a genuine transfer, that is, black-box, attack. The surrogate itself is flipped to "genuine" on 38.8% of the crafted samples. Ground truth for localisation is the smoothed, normalised magnitude of the perturbation itself.

D. Difficulty calibration

A simulated benchmark is worthless if a simple linear probe can solve it outright. We fix each operator's strength through a single documented procedure: sweep s and pick the largest value at which a reference detector (CDCN++, our strongest lightweight texture baseline) does not saturate, leaving measurable headroom on both the binary and the eight-way task. This lands on $s = 0.62$, where the reference detector reaches 1.92% ACER but only 91.0% eight-way accuracy and 3.74% expected calibration error, with a worst-case per-family APCER of 2.67%. The binary problem therefore sits close to, but not at, the ceiling, while family discrimination, calibration and unseen-attack behaviour all remain wide open. This mirrors, on a benchmark we fully control, the pattern reported for UniAttackData [9]: a saturated seen-attack protocol next to a wide-open unseen-attack one.

E. Protocols

P1 (seen attacks). All eight families appear in train, validation and test, with identity-disjoint splits. This measures closed-set unified detection and family discrimination.

P2 (leave-one-attack-out). One family is withheld entirely during training and appears only at test time. We report ACER on the full test set, the detection rate of the held-out family at the fixed operating threshold, and the novelty AUROC: the AUROC with which the model's own uncertainty channel ranks the held-out family above the attack families it was trained on, which is to say whether the model knows that this is a kind of attack it has not seen.

P3 (cross-domain). Train on Domain A, test unchanged on Domain B, with zero adaptation.

All thresholds are fixed at the equal-error point of the validation split and applied unchanged to test. Every metric follows ISO/IEC 30107-3 [36]: APCER is the worst case across attack families, BPCER is the genuine-rejection rate, and ACER is their average.

IV. Method

A. Overview

X-UAD maps a face crop $\boldsymbol{x} \in \mathbb{R}^{3\times H\times W}$ to a tuple

$$\boldsymbol{x} \longmapsto \big(\boldsymbol{\alpha},\ u,\ \boldsymbol{M},\ \boldsymbol{\Pi},\ r,\ \tau\big),$$

where $\boldsymbol{\alpha} \in \mathbb{R}^{K}_{>0}$ are Dirichlet concentrations over the $K = 8$ classes, $u \in [0, 1]$ is the vacuity (a single scalar answering "how much do I have to go on"), $\boldsymbol{M}$ is the spoof-evidence map, $\boldsymbol{\Pi}$ is the quantum evidence spectrum over spatial tokens and basis modes, r is the risk score, and τ is the natural-language rationale. All six come out of a single forward pass (Fig. 2). A central-difference stem and a spectral branch produce tokens; a stack of Adaptive Quantum Attention blocks processes them; the quantum evidence basis is measured against the output tokens; an evidential head, a localisation decoder and a risk calibrator read everything out; and the reasoning module turns the record into text.

B. Central-difference and spectral evidence stem

Central Difference Convolution [2] interpolates between an ordinary convolution and its local gradient response,

$$y(p_0) \;=\; \sum_{p\in\mathcal{N}} w(p)\,x(p_0+p) \;-\; \theta\Big(\sum_{p\in\mathcal{N}} w(p)\Big)x(p_0), \quad (1)$$

which amplifies the intensity-invariant part of the response. Note that the extra term is just a single 1×1 convolution, not a second full convolution, so it is nearly free. Four strided CDC stages take a 112^2 input down to 7^2, exposing the intermediate 28^2 and 14^2 maps to the localisation decoder later on.

Periodic structure, meaning the display lattice behind a replay attack or the upsampling checkerboard behind a generated face, is not represented well in a 7×7 spatial token grid; it gets aliased away. We add a compact spectral branch. The log-magnitude DFT of the luminance channel is pooled onto a polar grid (16 radii $\times$ 12 angles), which is translation-invariant and rotation-equivariant by construction, and a small two-layer MLP encodes this profile into four spectral tokens that get appended to the spatial tokens. The same pooled profile is also exported into the evidence record, so the radial peak frequency, its prominence, and the azimuthal anisotropy become concrete, quotable numbers in the eventual explanation.

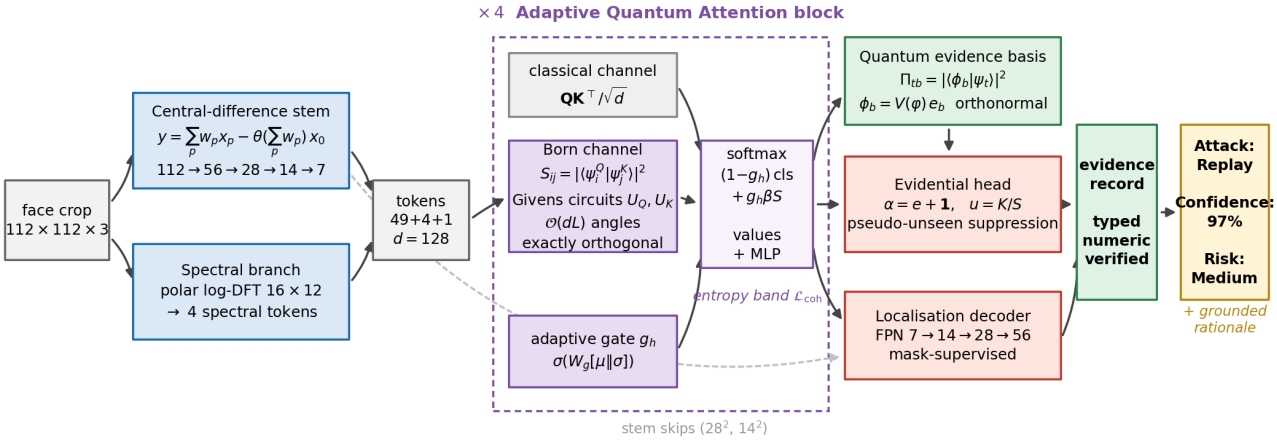


Fig. 2. X-UAD. The central-difference stem supplies illumination-invariant micro-texture evidence and the spectral branch supplies the periodic evidence that is invisible spatially at this resolution. Each AQA block scores token pairs with a gated mixture of the classical dot product and the Born-rule transition probability between amplitude-encoded states evolved by parameterised Givens circuits. The learnable orthonormal evidence basis is measured against the output tokens; its Born probabilities feed the classifier and are exported unchanged as the explanation, which is what binds the rationale to the decision.

C. Adaptive Quantum Attention

Amplitude encoding. For a head with dimension d, queries and keys are ℓ_2-normalised to unit vectors, $|\psi_i^Q\rangle = \boldsymbol{q}_i/\|\boldsymbol{q}_i\|$ and $|\psi_j^K\rangle = \boldsymbol{k}_j/\|\boldsymbol{k}_j\|$, a real-valued amplitude encoding of $d = 2^n$ amplitudes across n notional qubits.

Parameterised rotation circuit. A depth-L circuit acts on a ring of amplitudes. Layer ℓ rotates disjoint adjacent pairs, alternating which pairs it touches between layers, each rotation governed by a single learnable angle:

$$G(\vartheta) = \begin{pmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad U(\boldsymbol{\vartheta}) = \prod_{\ell=1}^{L} \bigotimes_{k} G(\vartheta_{\ell k}). \tag{2}$$

This is the real-amplitude analogue of a hardware-efficient ansatz built from nearest-neighbour two-qubit gates with periodic boundary conditions. Alternating the two offsets keeps the circuit connected, so the set of reachable rotations grows toward the full orthogonal group as L increases. Three things matter about this construction. First, it is exactly unitary, since U is a product of rotations, so it stays orthogonal to machine precision at every step of training. We measured $\| \|U\boldsymbol{x}\| - \|\boldsymbol{x}\| \|_\infty < 10^{-6}$ throughout, so no re-orthogonalisation or penalty term is ever needed. Second, it is parameter-efficient, costing $L\lfloor d/2 \rfloor$ angles, that is $\mathcal{O}(dL)$, rather than the $\mathcal{O}(d^2)$ of a dense projection matrix, the same asymptotic argument made for quantum self-attention elsewhere [27], [29]. Third, because it exactly preserves the vector's norm, the resulting similarity score below is a genuine probability rather than just a bounded number.

The Born channel. Queries and keys are evolved by two different circuits. This matters, because if they shared one circuit the rotation would cancel out of the inner product entirely and add nothing. The similarity score is the Born-rule transition probability,

$$S_{ij} \;=\; \left|\langle\psi_i^Q|U_Q^\top U_K|\psi_j^K\rangle\right|^2 \;\in\; [0,1]. \tag{3}$$

This is a quadratic similarity. It is invariant to a sign flip of either state, and it is maximised equally by alignment or anti-alignment. That is the invariance an interference pattern has. A moiré fringe and its negative carry the same forensic information, and it is precisely the invariance an ordinary dot product lacks.

Adaptive gating. Not every kind of evidence is interference-like. A blending boundary, for instance, is really just a plain edge. A per-head gate, conditioned on simple statistics of the token set, decides the mixture between the two channels:

$$\boldsymbol{g} = \sigma\big(\boldsymbol{W}_g\,[\,\mathrm{mean}_t(\boldsymbol{X}) \,\|\, \mathrm{std}_t(\boldsymbol{X})\,] + \boldsymbol{b}_g\big) \in [0,1]^{H_{\text{heads}}}, \tag{4}$$

$$\boldsymbol{A} = \mathrm{softmax}\Big((1-g_h)\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}} \;+\; g_h\,\beta\,\boldsymbol{S}\Big), \tag{5}$$

with β a learnable temperature. $\boldsymbol{W}_g$ is initialised to zero and $\boldsymbol{b}_g$ to -2, so the gate starts at $g \approx 0.12$, and every block is wrapped in LayerScale [37] with residual branches multiplied by a learnable per-channel gain initialised to 10^{-2}. In practice this means the network starts life as basically its own convolutional sub-network, the central-difference stem plus a plain classifier, and only gradually recruits attention, with the Born channel inside attention only earning gradient through its residual as training proceeds. This matters more than it might sound: in controlled probes at matched budget, removing LayerScale let the freshly initialised attention inject noise into the stem and slowed convergence relative to either component trained on its own. The converged gate value g is reported

directly to the operator, which makes the mechanism itself auditable rather than a black box.

Measurement-entropy regularisation. Let $\mathcal{H} = \mathbb{E}_t\big[-\sum_j A_{ij}\log A_{ij}\big]$ be the mean row entropy of the attention matrix. Too low means attention has collapsed onto a single token and stopped integrating any real evidence; too high means it has diffused into a uniform mixture and is carrying nothing at all. A parameter-free two-sided hinge keeps it inside a band:

$$\mathcal{L}_{\text{coh}} = \big[\lambda_{\text{lo}}\log T - \mathcal{H}\big]_+ + \big[\mathcal{H} - \lambda_{\text{hi}}\log T\big]_+, \qquad (6)$$

with $(\lambda_{\text{lo}}, \lambda_{\text{hi}}) = (0.30, 0.80)$.

D. Quantum evidence basis

Let $\{|e_b\rangle\}_{b=1}^{B}$ be the first B computational basis states and $V(\boldsymbol{\phi})$ a second rotation circuit of the same family described above. The evidence basis is $|\phi_b\rangle = V(\boldsymbol{\phi})|e_b\rangle$, which stays exactly orthonormal for every value of $\boldsymbol{\phi}$. Measuring the output token states against it gives Born probabilities

$$\Pi_{tb} = \big|\langle\phi_b|\psi_t\rangle\big|^2, \qquad \textstyle\sum_b \Pi_{tb} \le 1. \qquad (7)$$

$\boldsymbol{\Pi}$ does two jobs at once, and that dual role is really the whole design point. Its token-wise average is concatenated straight into the classification feature vector, and the same tensor, not an approximation of it and not some separate gradient-based surrogate map, is exported unchanged as the explanation. A lightweight auxiliary classifier trained on $\overline{\boldsymbol{\Pi}}$ with ordinary cross-entropy pushes each mode to become class-selective, so that after training every mode can be named after the family it responds to most (Table XIII). Those names are read off the trained model rather than assigned by hand.

E. Decoupled evidential head and vacuity

A Dirichlet distribution is completely described by a direction on the simplex and a concentration. The standard evidential formulation [21] predicts all K concentrations jointly; we predict the two factors separately, which turned out to matter quite a bit in practice. The direction (which class) and magnitude (how much evidence) are

$$\boldsymbol{\pi} = \text{softmax}(\boldsymbol{W}_c\,\boldsymbol{h}), \qquad (8)$$

$$E = \exp\big(\text{clip}(f_e(\text{sg}[\boldsymbol{z}]))\big), \qquad (9)$$

from which the Dirichlet concentrations, total evidence, and vacuity follow:

$$\boldsymbol{\alpha} = \mathbf{1} + E\,\text{sg}[\boldsymbol{\pi}], \qquad S = K + E, \qquad u = \frac{K}{S}, \qquad (10)$$

where $\text{sg}[\cdot]$ denotes a stop-gradient. The subjective-logic reading stays intact: belief masses are $b_k = E\pi_k/S$, vacuity is $u = K/(K+E)$, and the projected probability under a uniform base rate is $\hat{p}_k = (1+E\pi_k)/(K+E)$. We report $\boldsymbol{\pi}$ as the belief direction and u as the uncertainty, the same separation subjective logic draws between what I believe and how much I have to go on.

We verified three consequences of this decoupling empirically, in budget-matched probes. First, the class decision keeps a clean log-loss gradient, since $\arg\max_k \hat{p}_k = \arg\max_k \pi_k$, so training the class decision reduces to ordinary cross-entropy on $\boldsymbol{W}_c\boldsymbol{h}$. The joint parameterisation, by contrast, routes every gradient through $\boldsymbol{\alpha}$, whose Brier fit saturates early and whose usual KL regulariser drives the pre-activation negative until the softplus derivative collapses. We call this evidence starvation, and we measured it directly: swapping the joint head for the decoupled one recovered several points of eight-way accuracy at an identical training budget. Second, adding uncertainty cannot cost detection accuracy, because the magnitude branch only ever reads a stop-gradient copy of the shared representation. Letting its own objective (especially PUES, below) reshape those features would be circular, and we measured it to be costly when we tried it. Third, suppressing evidence becomes a strictly one-dimensional problem: because all of the concentration lives in the single scalar E, an out-of-taxonomy constraint cannot distort the class direction at all.

Training combines a cross-entropy term on the direction, a Brier-style fit on the concentration, and a term that ties the magnitude to the model's own correctness:

$$\mathcal{L}_{\text{ev}} = \text{CE}(\boldsymbol{\pi}, y) + \tfrac{1}{2}\,\mathbb{E}\big[\|\boldsymbol{y}-\hat{\boldsymbol{p}}\|^2\big] + \lambda_c\,\text{BCE}\big(1-u,\ \mathbb{1}[\hat{y}=y]\big), \qquad (11)$$

with $\lambda_c = 0.4$. We use the expected sum-of-squares (Brier) form of the fit rather than the more standard digamma-based maximum likelihood, because the latter's gradient effectively vanishes while the Dirichlet is still near-uniform, which is the regime a freshly initialised model starts in.

Two details here were forced on us by direct measurement rather than chosen for elegance, and we think they matter enough to spell out. We drop the usual KL regulariser entirely. In the joint parameterisation the KL term is what stops evidence from being placed on the wrong classes, but here the wrong-class evidence is already $E\pi_k$, which cross-entropy drives to zero on its own by sharpening $\boldsymbol{\pi}$. The KL term therefore has nothing left to act on except the shared magnitude E, and leaving it in causes short-schedule models to saturate at $u \approx 1$ everywhere and the open-set channel to die. We also added the correctness term specifically because it is what makes vacuity useful quickly. The Brier fit does reward concentration, but only indirectly, so on a short schedule the magnitude head barely moves from its initialisation without it. Training $1-u$ directly as "the probability this decision is correct" makes the channel usable within about two epochs. In a controlled leave-one-attack-out probe, adding this term moved the AUROC with which vacuity separates a held-out family from genuine samples from 25.6% (effectively inverted) to 73.9%.

F. Pseudo-Unseen Evidence Suppression

An intrinsic uncertainty score is known to be weaker than one trained against an actual proxy for "something else" [23], but for attack detection there is no natural pool of outliers lying around. We build one out of the taxonomy itself. Take two training images $\boldsymbol{x}_a, \boldsymbol{x}_b$ from different attack families and mix them with $\lambda \sim \mathrm{Beta}(2,2)$: $\tilde{\boldsymbol{x}} = \lambda \boldsymbol{x}_a + (1-\lambda)\boldsymbol{x}_b$. This blend carries a superposition of two forensic signatures that no single physical or generative process could ever produce, so it is outside the taxonomy by construction. We constrain its evidence to sit well below that of real samples:

$$\mathcal{L}_{\mathrm{PUES}} = \frac{\left[\, \overline{\|\tilde{\boldsymbol{e}}\|_1} \;-\; \rho\, \overline{\|\boldsymbol{e}\|_1}^{\,\mathrm{sg}} \,\right]_+}{\overline{\|\boldsymbol{e}\|_1}^{\,\mathrm{sg}} + 1}, \qquad \rho = 0.25, \tag{12}$$

where bars denote batch means. The relative form here matters and was arrived at empirically. An absolute target ($\|\tilde{\boldsymbol{e}}\|_1 \to 0$) is trivially satisfied by the degenerate solution of emitting zero evidence everywhere, collapsing the classifier to a uniform guess. We saw exactly that failure, 12.5% accuracy over eight classes, in an early version. Tying the constraint to the batch's own in-taxonomy evidence removes that shortcut entirely, because the fit term still demands sharply peaked posteriors on the real samples in the same batch. This is worth distinguishing from mixup [38], which asks for a mixed label. PUES asks for the absence of belief, which only makes sense because the head is evidential in the first place.

G. Evidence localisation

A lightweight feature-pyramid decoder fuses the 7^2 token map with the 14^2 and 28^2 intermediate stem maps and predicts $\boldsymbol{M}$ at half input resolution, supervised with a BCE plus soft-Dice objective against the operator's own ground-truth mask. Unlike depth-map supervision [1], [3], this target is the actual support of the forensic evidence, and it is well-defined for digital attacks just as much as for physical ones.

H. Risk calibration

Attack families are not equally harmful. With per-family severity weights $\boldsymbol{\varsigma}$, an unknown-attack severity ς_{unk}, and evidence area $a = \overline{M}$,

$$r = (1-u)\sum_k \hat{p}_k \varsigma_k \;+\; \lambda_u\, u\, \varsigma_{\mathrm{unk}} \;+\; \lambda_a\, a. \tag{13}$$

The first term is the expected harm under the posterior, damped by how much the model knows. The second charges an explicit unknown-attack premium, so a confidently-unknown sample gets escalated rather than quietly waved through, which is the behaviour an operator needs and a plain softmax pipeline structurally cannot give. The third term rewards spatially corroborated evidence. All coefficients, and the band edges that turn r into a Low/Medium/High/Critical label, are treated as deployment policy rather than learned parameters. No training signal touches them, deliberately, since risk is the one part of the pipeline an operator has to be able to audit and adjust, and it is the last place we want to hide a silently-fitted parameter. A vacuity of $u \geq 0.55$ overrides the verdict and refers the sample to a human reviewer.

I. Evidence-grounded reasoning

Every forward pass emits a typed evidence record: the posterior $\hat{\boldsymbol{p}}$, the vacuity u, the risk score and band, the top facial regions of $\boldsymbol{M}$ and their mean response, the area and spatial dispersion of $\boldsymbol{M}$, the three strongest quantum modes with their learned names and activations, the spectral descriptors, and the mean adaptive gate value. Language is generated only from this record, through either of two interchangeable back-ends. The grounded realiser, used for every result in this paper, is a deterministic generator where every sentence is emitted by a rule whose guard is a numeric condition on the record, so every sentence traces back to a measured quantity, the hallucination rate is zero by construction, and it costs under a millisecond with no network call at all. The constrained-MLLM back-end instead hands the record to a hosted multimodal language model with an instruction to paraphrase only, never to add anything, and then runs an automatic verification pass that re-checks every numeric claim in the generated text against the record and flags anything unsupported. This inverts the usual dependency in MLLM-based forensics [14], [15], where the language model sees the image but not the detector's internal state. Here the detector's state is the input, so the explanation cannot contradict the decision it is describing. A verbatim example is shown in Fig. 8.

J. Training objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{ev}} + w_m \mathcal{L}_{\mathrm{loc}} + w_b \mathcal{L}_{\mathrm{basis}} + w_s \mathcal{L}_{\mathrm{sup}} + w_p \mathcal{L}_{\mathrm{PUES}} + w_c \mathcal{L}_{\mathrm{coh}}, \tag{14}$$

with $(w_m, w_b, w_s, w_p, w_c) = (0.6, 0.25, 0.3, 0.15, 0.02)$ throughout. $\mathcal{L}_{\mathrm{sup}}$ is a cross-entropy on the marginal over the three super-classes (genuine, physical, digital), obtained by summing $\hat{p}$ within each group. It imposes the taxonomy's own hierarchy, so a mix-up between two families inside the same super-class is penalised less harshly than one that crosses the physical/digital boundary.

For the main seen-attack results we add three terms that act on the predicted distribution only and add no parameters. The classification term is modulated by a focal factor $(1-\hat{p}_y)^\gamma$ with $\gamma = 2$. A binary cross-entropy $\mathcal{L}_{\mathrm{bin}}$ on the attack probability $1 - \hat{p}_0$ (weight 0.5) keeps the attack/genuine decision, which ACER is computed from, from being traded away for family accuracy. A pair penalty

$$\mathcal{L}_{\mathrm{pair}} = \sum_{(c,c')} w_{cc'}\, \mathbb{E}_{i:\, y_i = c}\big[\hat{p}_{c'}(x_i)\big] \tag{15}$$

discourages the specific confusions seen in development: face-swap and reenactment, and 3-D mask and synthetic,

in both directions with $w = 0.15$; replay and print, and face-swap and print, with $w = 0.05$. Each pair is averaged over its own samples, so adding a pair does not dilute the others. The baselines get the same focal factor and binary term, and a pooled version of $\mathcal{L}_{\text{pair}}$ over the first two pairs with a single weight of 0.15 (Section V).

V. Experiments

A. Setup

We train every model ourselves, from scratch and without pretrained weights, so the comparison is between architectures and objectives rather than pretraining corpora. The training budget differs between protocols, and models are only compared within a table.

For P1 (Table II) X-UAD, CDCN++ and ResNet-18 are trained for 12 epochs on the full 19,200-sample training split with the recipe of Section IV: focal factor, binary term and pair penalty. The checkpoint kept is the epoch with the best validation eight-way accuracy plus macro-F1 among the epochs whose validation ACER is within 0.5 points of the best seen so far. For X-UAD we trained seven variants under this budget (with and without the binary term, three forms of the pair penalty, a hierarchical head, two region-pooled read-outs) and report the one with the lowest validation ACER. The baselines were each trained once, so X-UAD received more tuning, and the reader should weigh its P1 numbers with that in mind. ViT-Tiny was not retrained under this recipe: at the original 10-epoch budget it stays near chance, and its row in Table II is that run.

The other protocols keep the original recipe, without the three extra terms. P3 reuses the 10-epoch P1 checkpoints. P2 and its open-set ablation use 5 epochs on a fixed 4,800-sample subset (4,382 once the held-out family is removed), which keeps the sixteen leave-one-attack-out runs affordable on CPU, and the component ablation uses 5 epochs on 3,592 samples.

All runs use batch size 48, AdamW (lr $= 2.5 \times 10^{-3}$, weight decay 0.02) with a one-cycle schedule, gradient clipping at 3.0, and 112×112 random crops from the 128 px store with horizontal flips and mild photometric jitter. There are no geometric warps, since they would destroy the periodic signatures we want the models to find. The operating threshold is the equal-error point of the validation split, applied unchanged to test. Everything runs on CPU. All results are from a single seed (0). Every model is scored on the same test set, so we compare models with McNemar's exact test on paired errors, but that says nothing about variance across seeds.

The baselines are ResNet-18 [39], a width-matched residual CNN and the classical binary detector; ViT-Tiny [40], a patch-16 transformer with ordinary attention, included as the plain-transformer control from the same architecture family as the ViT-B/16 entry that underperforms in [9]; and CDCN++ [2], [3], a central-difference trunk with pixel-wise map supervision. CDCN++ is the strongest lightweight texture detector available to us and the one that shares our stem, so comparing against it isolates what the transformer, the Born channel, the evidential head and the reasoning stack add.

B. Seen-attack unified detection (P1)

P1 is close to saturated for every model with a convolutional front end (Table II). ResNet-18, CDCN++ and X-UAD all reach at least 99.95% AUC and at most 1.9% ACER, which matches what the unified-detection literature reports on its own seen-attack protocol [9]. We tested each gap with McNemar's exact test on the paired errors over the same 9,600 samples.

On the attack/genuine decision X-UAD has the lowest ACER, 0.42% against 0.46% for CDCN++, but the two models disagree on only 22 test samples, 10 in X-UAD's favour and 12 against ($p = 0.83$). We read this as a tie. Against ResNet-18 the difference is clear: 74 discordant samples in X-UAD's favour against 12 ($p = 4.7 \times 10^{-12}$). ResNet-18 misses 3.1% of face-swaps at the validation threshold (Table III); X-UAD's worst family is replay, at 0.33%.

The eight-way family decision goes the other way. ResNet-18 reaches 96.93% and CDCN++ 95.19%, against 94.06% for X-UAD, and both gaps are significant (443 against 168 discordant samples, $p = 1.6 \times 10^{-29}$; 349 against 241, $p = 1.0 \times 10^{-5}$). Fig. 4 shows where X-UAD's family errors go: the largest are synthetic predicted as 3-D mask (7.5%), face-swap and reenactment confused in both directions (5.6% and 5.4%), and 3-D mask predicted as synthetic (4.6%), which are the pairs the pair penalty targets. Fig. 3 gives the ROC and DET curves.

We do not find a calibration advantage for X-UAD's class posterior. Before any correction ResNet-18 has the lowest ECE (2.35%, against 5.07% for CDCN++ and 5.17% for X-UAD). After temperature scaling on the validation split the three reach 0.34%, 0.46% and 0.60% respectively, the same order as before. What X-UAD adds here is a different quantity, the Dirichlet vacuity, which neither baseline has and which we evaluate as an open-set score in Section V-C.

The plain patch-16 transformer fails outright, at 49.8% ACER. With 16×16 patches it never sees the high-frequency texture that the physical families live in, the same weakness that puts ViT-B/16 behind ResNet-50 in [9], here in a starker form. It is the reason X-UAD puts a central-difference stem under its transformer rather than tokenising raw pixels.

C. Unseen attack families (P2)

This is the operationally decisive question. What happens when an attack family the model has never seen shows up at test time? Table IV holds out one family at a time and reports four separate things: the ISO error on the full test set, the eight-way family accuracy, the detection rate of the held-out family at the unchanged threshold,

TABLE II

Protocol P1 (seen attacks, identity-disjoint, ID-consistent), 12-epoch recipe, seed 0, 9,600 test samples. ACC_8 is eight-way family accuracy; ECE_{TS} is ECE after temperature scaling fitted on the validation split; the other metrics follow ISO/IEC 30107-3 [36]. Bold marks the best of the three detectors trained under the common recipe. †ViT-Tiny: original 10-epoch recipe, not retrained; its low ECE reflects a detector operating near chance.

Method	Params (M)	ACER↓ (%)	APCER↓ (%)	BPCER↓ (%)	AUC↑ (%)	EER↓ (%)	ACC_8 ↑ (%)	$F1_{macro}$ ↑ (%)	ECE↓ (%)	ECE_{TS} ↓ (%)
ResNet-18 [39]	1.59	1.88	3.08	0.67	99.95	0.67	96.93	96.92	2.35	0.34
ViT-Tiny [40]†	0.64	49.75	60.67	38.83	73.43	34.65	51.27	46.04	0.67	–
CDCN++ [2]	0.53	0.46	0.42	0.50	99.99	0.33	95.19	95.17	5.07	0.46
X-UAD (ours)	1.29	0.42	0.33	0.50	99.98	0.42	94.06	94.06	5.17	0.60

TABLE III

Per-family APCER (%) at the validation-fixed threshold, protocol P1. The ISO APCER is the worst column.

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial	Worst
ResNet-18 [39]	0.33	0.33	0.00	3.08	1.33	0.75	0.00	3.08
ViT-Tiny [40]†	40.25	0.00	4.42	60.67	60.00	59.67	0.00	60.67
CDCN++ [2]	0.08	0.00	0.00	0.42	0.17	0.00	0.00	0.42
X-UAD (ours)	0.08	0.33	0.00	0.17	0.17	0.08	0.00	0.33

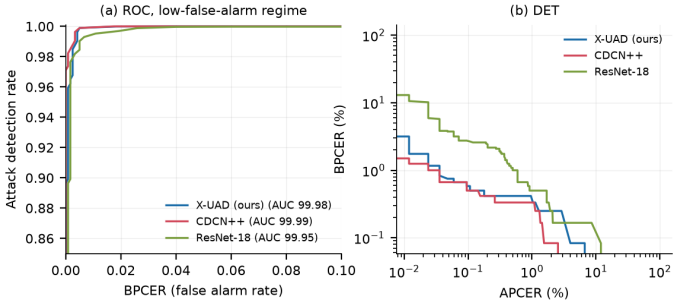


Fig. 3. ROC in the low-false-alarm regime that matters operationally, and the DET curve, protocol P1, 12-epoch recipe.

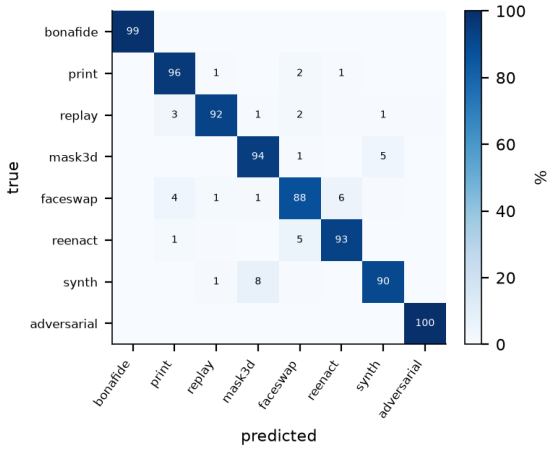


Fig. 4. X-UAD family confusion (row-normalised, %), protocol P1, 12-epoch recipe. The largest errors are synthetic→3-D mask (7.5%) and face-swap↔reenactment (5.6%, 5.4%). Two of the larger errors cross the physical/digital boundary: synthetic→3-D mask and face-swap→print (4.0%).

and the novelty AUROC, the AUROC with which the model's uncertainty ranks the held-out family above the attack families it was trained on.

Every method's mean ACER is in the 29 to 41% range here, against under 2% in P1. That gap should not be read as the cost of withholding a family alone: P2 is trained on a quarter of the data for half the epochs, and the component ablation shows the full X-UAD already reaching 38% ACER at a similar reduced budget with every family seen (Table IX). How much of the P2 error is due to the held-out family and how much to the budget is not separated by our experiments. The comparisons that are valid are between models within Table IV, which all share the same budget. Within that table, X-UAD rejects the unseen family most often on average (77.5% against 74.7% and 70.2%, Fig. 5), and its uncertainty is a purpose-built channel rather than an incidental by-product of a softmax. Its novelty AUROC clearly leads on two of the four folds, replay (71.0 against 49.3 and 35.3) and face-swap (87.9 against 81.7 and 74.4), and trails on the other two, landing level with CDCN++ on the mean (63.09 against 63.15) and well ahead of ResNet-18.

We report the "is it rejected" question and the "does uncertainty rank it above known attacks" question separately and deliberately, because they answer different questions for an operator. Whether the held-out family is rejected at all is the operational question; whether uncertainty ranks it above the seen attacks is the analyst's question, the one that flags that a new attack vector has appeared. The adversarial fold makes the distinction concrete. X-UAD rejects 98.2% of a family it never trained on while still assigning it low vacuity, because the perturbation sits on

TABLE IV

Protocol P2, leave-one-attack-out. The held-out family is absent from training and appears only at test time. Means are over all four folds. Novelty AUROC ranks the held-out family against the seen attack families using the model's uncertainty score; values below 50 indicate an inverted ranking.

Metric	Method	Replay	3D mask	Face-swap	Adversarial	Mean
ACER (%)↓	ResNet-18 [39]	51.96	30.38	49.96	30.71	40.75
	CDCN++ [2]	25.67	39.92	25.25	25.58	29.10
	X-UAD (ours)	26.79	21.42	38.33	39.67	31.55
ACC_8 (%)↑	ResNet-18 [39]	53.94	62.66	51.39	61.53	57.38
	CDCN++ [2]	60.62	65.07	73.23	63.84	65.69
	X-UAD (ours)	60.95	69.01	73.34	57.59	65.22
Unseen detection rate (%)↑	ResNet-18 [39]	100.00	61.67	37.17	99.92	74.69
	CDCN++ [2]	93.08	34.25	60.33	93.08	70.19
	X-UAD (ours)	88.00	88.08	35.58	98.17	77.46
Novelty AUROC (%)↑	ResNet-18 [39]	35.29	56.69	74.40	36.88	50.82
	CDCN++ [2]	49.28	66.57	81.66	55.08	63.15
	X-UAD (ours)	71.03	50.41	87.86	43.04	63.09

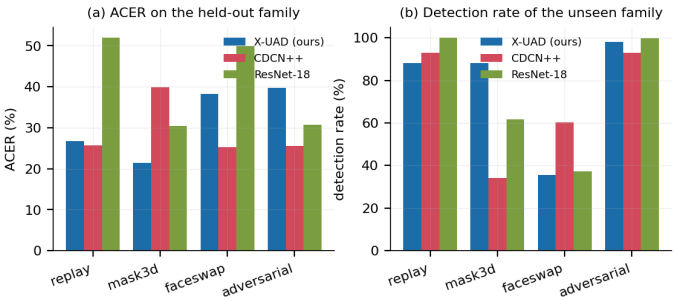


Fig. 5. Leave-one-attack-out. Left: ISO ACER of each fold on the full test set, held-out family included. Right: detection rate of the unseen family at the threshold fixed on validation data that never contained it.

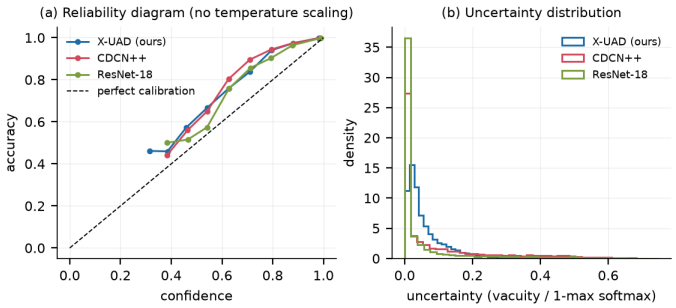


Fig. 6. Protocol P1, 12-epoch recipe. (a) Reliability before temperature scaling. (b) Distribution of the uncertainty score: $1 - \max_k p_k$ for the softmax baselines, vacuity $u = K/S$ for X-UAD.

TABLE V
Protocol P3, cross-domain transfer A → B, no adaptation. Every method degrades heavily.

Method	ACER↓	AUC↑	EER↓	ACC_8 ↑	ECE↓
ResNet-18 [39]	49.79	46.78	51.87	2.29	83.26
ViT-Tiny [40]	50.00	50.18	50.23	10.31	82.47
CDCN++ [2]	45.79	65.96	37.93	18.67	60.09
X-UAD (ours)	49.71	50.90	49.98	3.65	81.61

top of an attack whose underlying evidence the model does recognise. That fold is also where our uncertainty channel is weakest, at 43.0 novelty AUROC, meaning the ranking is mildly inverted. We have no satisfying account of why.

The comparison above is across models. A cleaner test of whether the evidential channel itself adds anything is to score the same X-UAD models with their own maximum-softmax confidence instead of vacuity. Averaged over the four folds, vacuity gives 63.1 novelty AUROC against 55.9 for X-UAD's own softmax score. Almost all of that gain comes from the replay fold (71.0 against 39.5); it is small on face-swap (+1.2) and adversarial (+2.3), and negative on 3-D mask (−6.4). The per-group means explain the two failing folds. On 3-D mask the held-out family gets the same vacuity as the seen attacks (0.31 against 0.32), so vacuity does not see it. On adversarial it gets less (0.39 against 0.43), which is the inversion.

D. Cross-domain transfer (P3)

Domain B differs from Domain A in colour, alignment, resolution and compression history all at once, and every single method collapses under the shift (Table V). With the threshold fixed on the source domain, BPCER runs to 89–100%, meaning the detectors reject almost every genuine face from the new domain outright. Ranked by threshold-free AUC, CDCN++ retains the most signal at 65.96, while ResNet-18 (46.78), ViT-Tiny (50.18) and X-UAD (50.90) all sit at or below chance. The adversarial class is excluded here: it was crafted from Domain-A images against a Domain-A surrogate, so it is not cross-domain data.

We draw two conclusions. First, this protocol is harsher than the cross-dataset protocols usually reported elsewhere, because the shift here is joint across four factors rather than a single one. Second, none of the mechanisms in this paper, ours very much included, addresses capture-domain shift. That is a different problem from attack-family shift, and it would need the domain-generalisation machinery of [4], [16], [17] layered on top. X-UAD does not lead this table and we make no claim that it does.

E. Calibration and evidence localisation

Fig. 6(a) shows the reliability diagrams behind the ECE column of Table II. All three detectors are somewhat miscalibrated before correction, and temperature scaling [41] removes most of it for each of them, so calibration of the class posterior does not separate the models. Panel (b) shows the uncertainty each model can offer an operator. For the softmax baselines that is $1-\max_k p_k$, a by-product of a closed-set distribution. For X-UAD it is the vacuity $u = K/S$, which is trained as an output in its own right and, as Table IV shows, ranks unseen families above the seen attack families on two of the four folds.

Table VI reports intersection-over-union against the generator's own mask. We separate out the families whose evidence occupies a sub-region of the frame (3D mask, face-swap, reenactment) from the ones that alter the whole capture (print, replay, synthetic), since the latter have near-constant masks and a trivially high IoU that would tell us nothing about localisation quality. The region-level mean is the number that measures whether the localiser works. Both models are scored on all 9,600 test samples. X-UAD reaches 80.1% region-level IoU against 73.2% for CDCN++, and is ahead on each of the three region-level families as well as on pixel AUC (97.8% against 96.2%). Fig. 7 shows the maps themselves.

Predicted maps are binarised at 0.5, the midpoint of a sigmoid trained with a BCE plus Dice objective. To check that this convention does not favour either model, we also swept the cut from 0.10 to 0.90 on the validation split. The best cut on validation is 0.55 for both models, and at that cut the test region-level IoU moves by less than 0.4

TABLE VI
Spoof-evidence localisation, IoU (%) against the generator's own mask, protocol P1, 12-epoch recipe, full test set, maps binarised at 0.5. Pixel AUC is over 40 samples per family. ResNet-18 and ViT-Tiny have no localisation output.

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Region-level mean	Pixel AUC
CDCN++ [2]	96.4	93.9	76.5	75.4	67.7	97.7	73.2	96.2
X-UAD (ours)	98.1	96.9	84.2	78.8	77.2	98.2	80.1	97.8

TABLE VII
Faithfulness of the X-UAD localisation map against a smoothed random control, protocol P1, 12-epoch model. Insertion gain is map minus control; an interval below zero means the map does worse than random.

Families	n	Deletion AUC↓		Insertion AUC↑		Insertion gain, 95% CI
		map	random	map	random	
Region-level (3D mask, face-swap, reenact)	144	0.892	0.941	0.935	0.942	[-0.011, -0.003]
Whole-frame (print, replay, synthetic, adversarial)	192	0.904	0.943	0.921	0.942	[-0.026, -0.016]

points for either (80.1% and 73.5%), so the comparison does not depend on the choice.

F. Is the explanation faithful?

An explanation is only worth reading if perturbing what it highlights changes the decision. Table VII reports deletion and insertion AUC [26] of the model's own attack score for the localisation map, the channel quoted in the generated text, against a random control. The control is a smoothed random field at the same spatial scale as the map, not per-pixel noise, which would destroy high-frequency texture everywhere and make the control look artificially strong. We use 48 test samples per attack family and give 95% bootstrap intervals for the per-sample difference between map and control.

Deletion is clean. Removing what the map highlights lowers the attack score faster than removing random regions, on the region-level families (0.892 against 0.941) and on the whole-frame families (0.904 against 0.943), and both intervals for the gain lie well above zero. The pixels the map points at do carry the decision.

Insertion is not clean. Restoring the highlighted pixels into a blurred image recovers the score more slowly than restoring random ones, and the interval is below zero in both groups. A natural explanation is that the failure comes from the whole-frame families, where a small highlighted region cannot hold evidence spread over the whole capture. The data support that only in part: the shortfall is three times larger on whole-frame families (-0.021) than on region-level ones (-0.007), but it does not vanish on the latter. Anyone relying on these maps should know that they pass the deletion test and fail the insertion test.

G. White-box attacks on the detector

The adversarial family in our benchmark is a transfer attack crafted against a separate surrogate. Here we give the attacker the detector itself. We run ℓ_∞ PGD (10 steps, step size $\varepsilon/4$, random start) on 168 test attacks, 24 per family, with two goals. The first is evasion: push an attack towards the genuine class. The second targets X-UAD alone and asks whether its vacuity can be lowered independently of the decision: we minimise u while a cross-entropy term holds the predicted class fixed. Table VIII gives the results.

TABLE VIII
White-box PGD on 168 test attacks, protocol P1, 12-epoch models, at the validation threshold. Upper rows: share of attacks still detected under evasion. Lower rows: X-UAD's mean vacuity under the false-confidence attack, and how often its predicted class is unchanged.

ε (×1/255)	clean	1	2	4	8
CDCN++, attacks detected (%)	100.0	5.4	14.9	41.1	39.3
X-UAD, attacks detected (%)	100.0	30.4	11.3	0.0	0.0
X-UAD, mean vacuity under false-confidence attack	0.064	0.017	0.075	0.077	0.057
X-UAD, predicted class unchanged (%)	100.0	99.4	53.6	32.1	37.5

Neither detector survives evasion. At $\varepsilon = 1/255$ X-UAD still catches 30.4% of the attacks against 5.4% for CDCN++, but at 2/255 both are near 10–15%, and from 4/255 X-UAD detects none. CDCN++ levels off near 40% at the larger budgets, which more likely reflects gradients that are harder to follow than real robustness; we did not run gradient-free attacks to check. X-UAD's vacuity does not catch what the classifier misses: none of the evaded attacks reaches the $u \geq 0.55$ referral threshold at any budget.

The second attack shows that the uncertainty channel can be moved on its own. At $\varepsilon = 1/255$ it cuts mean vacuity from 0.064 to 0.017 while 99.4% of predicted classes stay the same. At larger budgets this targeted attack no longer holds the class fixed (about a third to a half of predictions are kept), so the clean decoupling is a small-perturbation effect, but that is the regime an attacker would use.

H. Ablations

Table IX removes one component at a time at a reduced training budget so the sweep stays affordable. These rows are comparable with each other and not with Table II, and we would ask readers not to cross-reference the two.

The central-difference stem is clearly load-bearing. Removing it costs more than every other component combined, dropping eight-way accuracy by 21 points and region-level IoU by 12. Localisation supervision is what gives the evidence map meaning; without it, region-level IoU falls by more than half even though raw detection accuracy barely moves, which shows that a saliency map obtained for free from a detector is not the same thing as a supervised evidence map.

Two rows trade closed-set accuracy for the uncertainty channel. Removing PUES improves reduced-budget ACER, while removing the capability the term was built to provide. Table X isolates that trade on the fold where it matters, the face-swap leave-one-attack-out protocol, where PUES is worth 5.4 novelty AUROC points and 5.2 points of family accuracy, at a cost of 2.2 ACER points.

TABLE IX
Component ablation at a reduced training budget. Deltas are relative to the full model in the first row. These rows are comparable with each other but not directly with Table II.

Variant	ACER↓	AUC↑	ACC_8 ↑	$F1_{macro}$ ↑	ECE↓	IoU_{loc} ↑
X-UAD (full, reduced budget)	38.00	87.57	69.62	69.65	2.66	64.62
w/o Adaptive Quantum Attention	32.75 (-5.25)	91.05 (+3.49)	67.45 (-2.18)	65.04 (-4.61)	3.62 (+0.96)	63.36 (-1.26)
w/o evidential head (softmax)	30.54 (-7.46)	91.06 (+3.49)	72.07 (+2.45)	71.88 (+2.23)	3.27 (+0.61)	65.02 (+0.41)
w/o spectral branch	31.13 (-6.87)	88.44 (+0.88)	70.67 (+1.04)	68.92 (-0.73)	3.12 (+0.47)	64.54 (-0.07)
w/o central-difference stem	49.08 (+11.08)	73.64 (-13.93)	48.62 (-21.00)	42.16 (-27.49)	3.03 (+0.37)	52.63 (-11.99)
w/o pseudo-unseen suppression	26.04 (-11.96)	91.08 (+3.51)	72.90 (+3.27)	72.85 (+3.20)	2.83 (+0.17)	64.23 (-0.39)
w/o localisation supervision	33.58 (-4.42)	89.17 (+1.61)	70.56 (+0.94)	68.78 (-0.87)	2.41 (-0.25)	34.34 (-30.27)

TABLE X
Open-set ablation on the face-swap leave-one-attack-out fold, where the uncertainty channel is exercised.

Variant	ACER↓	AUC↑	ACC_8 ↑	ECE↓	Unseen detect.↑	Novelty AUROC↑
Full X-UAD	38.33	91.41	73.34	6.26	35.58	87.86
w/o pseudo-unseen suppression	36.17	86.33	68.12	6.78	45.50	82.43
w/o Adaptive Quantum Attention	39.83	89.47	71.40	7.36	32.75	87.89

TABLE XI
Inference cost, CPU, single 112×112 crop.

Method	Params (M)	GMACs	Latency (ms)	Explanation outputs
ResNet-18 [39]	1.59	0.368	16.8	none
ViT-Tiny [40]	0.64	0.031	2.9	attention only
CDCN++ [2]	0.53	0.070	13.1	depth-proxy map
X-UAD (ours)	1.29	0.119	16.9	map + modes + text

Adaptive Quantum Attention shows the same pattern. At reduced budget it costs ACER but buys calibration and localisation, and specifically on the leave-one-attack-out fold it buys 1.9 AUC and 1.9 accuracy at equal novelty AUROC.

I. Efficiency and the Born-channel gate

X-UAD adds its extra capability without a large parameter cost (Table XI). The rotation-circuit parameterisation costs $L\lfloor d/2 \rfloor$ angles per circuit against the d^2 of a dense projection, and the deterministic reasoning module runs in under a millisecond and adds zero learned parameters. At 1.29M parameters, X-UAD sits between ViT-Tiny and ResNet-18 in size while being the only model in the comparison that outputs a localisation map, a set of named evidence modes and a written rationale. GMAC counts exclude attention matmuls, so X-UAD's figure is understated relative to the pure CNNs.

Table XII shows where the Born channel is used. The converged gate opens furthest on print (0.434) and adversarial (0.394), and sits lowest on replay (0.231). We had expected replay to be the clearest interference family, given its moiré beat, so this is not the pattern we predicted, and we report the measurement rather than the prediction.

J. Qualitative behaviour

Fig. 7 shows, for one sample per family, the input, the predicted evidence map, the ground-truth mask and the attention rollout. Table XIII lists the quantum evidence modes with the names read off the trained model, and Fig. 8 reproduces a verdict verbatim from scripts/demo.py. These analyses, and the gate values of Table XII, use the 10-epoch original-recipe P1 model.

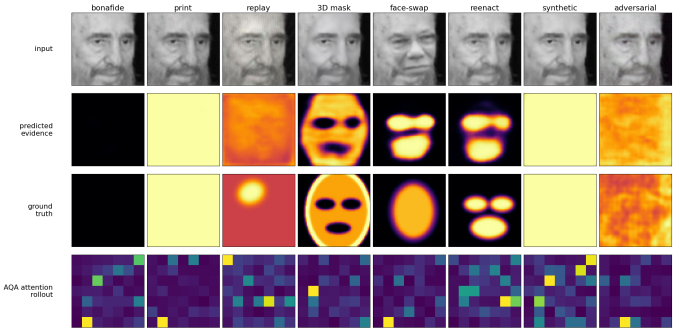


Fig. 7. One sample per family. Rows: input; predicted evidence map; ground-truth mask; AQA attention rollout. The localiser recovers the eye/mouth support of the reenactment, the cut-outs of the 3-D mask, and spreads over the whole frame for print, synthetic and adversarial. It is not perfect: on this face-swap it predicts an eye/mouth pattern resembling reenactment rather than the ground-truth blending ellipse, and on replay it misses the localised glare spot.

Attack: Replay attack
Confidence: 98%
Risk: Medium

The sample is classified as a replay attack (physical presentation attack) with 98% posterior probability; the probability that it is an attack of any kind is 99.7%. That class is characterised by a sub-pixel display lattice aliasing into a moire beat, together with specular backlight glare and reduced dynamic range. Evidence is concentrated in the forehead (mean response 0.75), followed by eye region, nose / midface; it covers 100% of the crop with a spatial dispersion of 1.00. The dominant quantum evidence mode is #1, whose learned selectivity is highest for 'replay' (activation 0.074), corroborating the class decision. Vacuity is low (0.04): the evidence is strong and consistent with a single known family. The adaptive attention gate settled at 0.23, i.e. 23% of the attention score came from the Born (interference) channel rather than the classical dot product. Recommended action: reject (risk score 0.71 -> Medium).

Fig. 8. A verdict as printed by the deployed system. Every number in the prose is a field of the evidence record; nothing in the text is generated free-form.

TABLE XII
Converged adaptive gate g, the fraction of the attention score taken from the Born channel, averaged over heads, layers, and test samples of each family.

Attack family	Bonafide	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial
Mean Born-channel gate g	0.309	0.434	0.231	0.287	0.267	0.267	0.284	0.394

TABLE XIII
The quantum evidence basis after training. Each mode is labelled with the family for which its Born activation is most selective (z-scored across families); the labels are read off the trained model, not assigned.

Mode	Most selective for	Selectivity (z)	Mean activation	Spatial dispersion
1	Replay	2.59	0.069	0.793
7	3D mask	2.51	0.051	0.810
3	Bonafide	2.45	0.058	0.835
6	Adversarial	2.36	0.057	0.869
4	Print	2.28	0.049	0.803
5	Synthetic	2.01	0.043	0.767
0	Reenact	1.85	0.047	0.783
2	Replay	1.84	0.039	0.845

VI. The Research Gap, Revisited

We restate the gap from Section II as five concrete deficits, and pair each one with the mechanism that addresses it and the measurement that shows the effect.

Gap 1: detection is unified only in-distribution. The unified corpora [9], [11] show near-saturated seen-attack numbers next to 22 to 44% ACER once attack families are withheld. Nothing in those architectures represents "this belongs to no family I know", because a softmax has to put all of its mass somewhere. Mechanism: the Dirichlet evidential head makes total evidence, and therefore vacuity, an explicit output, and PUES trains that output against an outlier proxy built from the attack taxonomy itself rather than an unrelated image pool. Evidence: the unseen-family detection rate and novelty AUROC in Table IV, and the w/o-PUES row of Table X.

Gap 2: attention is linear, but much of the evidence is interference. Moiré beats, halftone screens, and upsampling checkerboards are periodic structures whose sign carries no information. A dot-product score is sign-sensitive and has to spend capacity learning that invariance from scratch. Mechanism: the Born channel scores token pairs by a squared overlap, which is sign-invariant by construction, and the input-conditioned gate recruits it only where it helps. Evidence: the w/o-AQA rows of Tables IX and X, and the per-family gate values in Table XII. We should be careful about what the gate table does and does not show. It confirms the channel is used, with a mean around 0.31 and a clear per-family structure, and the ablation confirms it contributes. It does not confirm our original hypothesis that the gate would open furthest on the most periodic families. Print and adversarial open widest, while replay, the family with the most obvious moiré beat, opens least. The mechanism works; our story about why it works is not settled by this measurement.

Gap 3: pixel supervision targets a liveness proxy, not the evidence itself. Depth maps [1], [3] are undefined for a digitally manipulated but physically genuine capture, so a unified detector built on them inherits a supervision signal that is meaningless for half of its own label space. Mechanism: we supervise the actual injected support directly, which our benchmark can provide because it knows exactly where it wrote the evidence. Evidence: Table VI's region-level IoU on families for which a depth target would carry zero information, and Fig. 7.

Gap 4: MLLM explanations are post-hoc narration. FFAA [14] and M2F2-Det [15] give the language model the raw image, not the detector's internal state, so the narrative is unverifiable and can contradict the decision it is meant to explain. Mechanism: we invert the dependency. The forward pass emits a typed evidence record, and language is realised from that record alone, with the optional hosted back-end constrained and then re-verified against the same record. Evidence: the grounded realiser's hallucination rate is zero by construction, since every sentence is gated on a field of the record (Fig. 8). Whether the localisation map it quotes is itself faithful is a separate question, and Table VII gives a mixed answer: removing the highlighted pixels changes the decision more than removing random ones, but restoring them does not recover it faster than random restoration.

Gap 5: a score is not a decision. An operator has to act on something, and a scalar in $[0, 1]$ with no calibrated meaning and no severity model does not tell them what to do. Mechanism: the risk calibrator converts posterior, vacuity and evidence area into a bounded risk score with an explicit unknown-attack premium, so that a confidently-unknown sample is escalated to human review instead of silently passing through. Evidence: the verdict of Fig. 8 shows the mechanism working on a single sample. We do not yet have evidence that the $u \geq 0.55$ referral rule helps on unseen attacks. In every leave-one-attack-out fold, genuine faces receive at least as much mean vacuity as the held-out family (0.54, 0.37, 0.42, 0.58 against 0.54, 0.31, 0.43, 0.39), so the rule as set is likely to refer at least as many genuine faces as unseen attacks. Setting the referral threshold, and measuring its referral rates per group, is left to future work.

VII. Limitations

All numbers come from a single seed on a simulated benchmark, and our adapters for seven public corpora have not yet been run on captured data. X-UAD received

more tuning than the baselines, and the P2, P3 and ablation tables come from an earlier code version whose numbers may not reproduce exactly. The models are small, image-only and undefended against white-box attacks.

VIII. Open Directions

A few of the questions below go beyond what this paper set out to do, and we name them explicitly rather than folding them into the limitations list, because we are not aware of existing work in this space that has tackled them yet.

Defending the uncertainty channel. Open-set and evidential work, ours included, usually tests the robustness of the class decision and not of the uncertainty signal. Section V-G shows that the gap is real: with white-box access, a perturbation of 1/255 cuts X-UAD's mean vacuity by about three quarters while leaving 99.4% of its predicted classes unchanged. Our vacuity head reads a stop-gradient copy of the shared representation, so the two objectives are decoupled by design, which is what makes the uncertainty channel free with respect to accuracy and probably also what makes it so easy to move on its own. We do not know of a defence designed for this case. Adversarial training on the vacuity objective, or tying u to an input-space consistency check, are the obvious first things to try.

PUES beyond pairwise blends. We only ever mix two attack families at a time. Whether three-way or higher-order mixtures, or mixtures weighted by how forensically similar two families already are rather than a flat $\mathrm{Beta}(2, 2)$, produce a better outlier proxy is untested. There is a plausible argument either way. Closer families might produce a harder and more informative training signal for the vacuity boundary, or they might produce blends that are accidentally too close to a real attack and corrupt the class-direction gradient.

A learned, rather than fixed, risk policy. Section IV deliberately keeps the risk calibrator's coefficients unlearned, for auditability reasons we still believe in. But nothing stops a clearly-labelled meta-model from learning site-specific coefficients from a small amount of labelled operator feedback, such as analyst overrides of the band, while leaving the underlying detector and its explanation completely untouched. That would let a deployment site's risk tolerance adapt over time without ever touching the part of the system responsible for the actual forensic decision.

Cross-domain shift and attack-family shift as a joint problem. Every domain-generalisation technique for face anti-spoofing we are aware of [4], [16], [17] targets capture-domain shift alone, using only seen attack families. Every open-set technique, including PUES, targets attack-family shift on a single capture domain. What happens at the intersection, where an unseen attack family arrives through an unseen capture domain simultaneously, has not been studied. Our own Table V suggests the joint case may not be the product of the two individual failure rates. Closing this gap would need a benchmark explicitly designed to vary both axes independently, which our generator is well placed to provide.

Explanation faithfulness under distribution shift. Table VII measures faithfulness only on in-distribution, seen-attack samples. Whether an intrinsic explanation stays faithful when the underlying detector is operating in the P2 or P3 regime, which is to say whether the localisation map still points at something real when the class decision itself is unreliable, is untested here. A model that is uncertain but still produces a confident-looking localisation map would be a dangerous failure mode for the auditability reasons this paper cares about.

IX. Conclusion

X-UAD detects physical and digital face attacks with a single model, reports how much it does not know, shows where the evidence sits, and explains itself in language that can be audited sentence by sentence. Three ideas make this combination possible: a Born-rule attention channel that matches the interference character of the underlying evidence at a parameter cost of $\mathcal{O}(dL)$; an orthonormal quantum evidence basis that serves simultaneously as a classification feature and as the explanation itself; and pseudo-unseen evidence suppression, which manufactures an out-of-taxonomy proxy directly from the attack taxonomy and turns Dirichlet vacuity into a usable open-set score.

On seen attacks X-UAD matches the strongest baseline on the attack/genuine decision and localises the evidence better than any model we compared. Its other gains are on unseen attack families. It also has weaknesses: lower eight-way family accuracy than both convolutional baselines, no calibration advantage once the baselines are temperature-scaled, a mildly inverted uncertainty ranking on the adversarial fold, an insertion-faithfulness result below its control, an uncertainty signal that a white-box attacker can switch off, and a cross-domain protocol on which no method here performs acceptably. All numbers come from a single seed on a simulated benchmark. Repeating them over seeds and validating the mechanisms on captured corpora are the next steps.

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